Kasin Ito-Sato

Mechanical engineering student at Loyola Marymount University (BS, May 2027).
Design, build and test work, with measured results reported against predictions.

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Project · MECH Machine Design · Spring 2026

Rope Climbing Robot

A motorized robot that climbs along a rope to carry itself across a "river." After the arm design failed in testing, a screw-slider redesign crossed 38 ft in 5.5 minutes.

Predicted	0.134 ft/s
Measured	**0.115 ft/s**

About 86% of predicted. The gap likely comes from slip on the rope and drivetrain friction.

My role. Four-person team. Built, assembled and tested the 3-stage spur gear drivetrain (64T/32T/16T), and helped rebuild the mechanism as a constrained screw slider.

Concept Selection · SolidWorks · Motion Analysis · Kinematic Analysis · Laser Cutting · 3D Printing · Gear Trains



Project · MECH 3100 Fluid Mechanics · Fall 2025

Small Wind Turbine Design

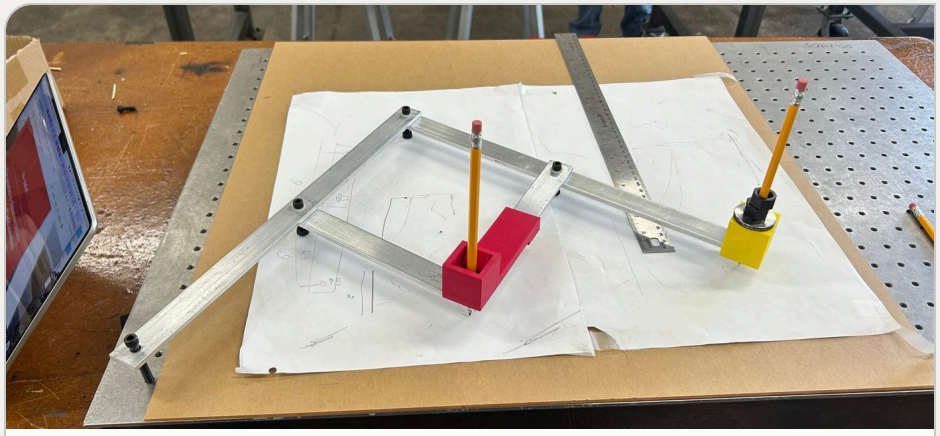
Designed, 3D printed and wind-tunnel tested a four-blade turbine, then compared measured power against hand calculations and CFD.

Wind tunnel	3 speeds, 208 to 422 RPM
Measured power	**up to 6.2 mW**

Traced why output fell short of the 0.05 to 0.33 W ideal model: motor friction, drag and blade imbalance.

My role. Three-person team. Airfoil selection, hand calculations, SolidWorks model and flow simulation, wind tunnel testing.

SolidWorks · SolidWorks Flow Simulation · 3D Printing · Wind Tunnel Testing · Airfoil Selection · Hand Calculations



Project · MECH 2100 Design & Manufacturing · Fall 2024

2x Magnifying Pantograph

Designed and built an aluminum linkage that enlarges a drawing by exactly 2.0x, with GD&T bar drawings and a tolerance analysis.

Target	2.0x
Measured	**1.98x to 2.0x**

Exactly 2.0x on a 2 in line, never over 2.0x on a 2 x 2 in box.

My role. Two-person team. Linkage layout, SolidWorks model, GD&T drawings, machining and testing.

Linkage Design · SolidWorks · GD&T · Milling · Bandsaw · 3D Printing · Tolerance Analysis



Punahou School · Engineering III

Go-Kart

A four-person team build of a go-kart in six weeks, from a kit and 2 x 1 in aluminum rectangular tubing: research, sketches, CAD, chassis, drive, steering and brakes.

- Reworked chassis bars when the brackets had off-set holes, and made and welded our own turning mechanism when the kit parts were too long.
- Took the 3D printed steering wheel cap through three prints: the first came out as two weak pieces, the second broke from too little infill, the third held.

My role. Designed the drive's axle mount bearing, brake disc mount, sprocket mount and spacers, and the steering wheel cap body.

CAD · Chassis Fabrication · Welding · Drilling · 3D Printing · Drive and Steering Systems

More projects at kasinitosato.github.io/projects

SAE Aero Design Regular Class Competition
Senior capstone designing a two-motor electric RC cargo plane that carries 2 L bottles for the 2027 SAE Aero Design Regular Class.

Toggle Clamp + Left Support DFM
Modeled a full toggle clamp assembly in SolidWorks, then redesigned its left support twice, as a machined part and as sheet metal, with full GD&T.

Hydraulic System Design for a School in Waslala, Nicaragua
Designed the potable water system for a rural school and sized its pump with a MATLAB pipe network solver. Required pump head was 23.4 m.

Heat Sink and Forced-Convection Testing
Built an Arduino thermocouple setup to measure how a fin array and moving air pull heat from a heated block and cylinder.